Automated Stone Counting System for Embroidered Textile Fabrics Using Machine Learning

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***Abstract*—Stone detection in fabric is an essential process in textile manufacturing because it ensures quality of products and minimizes the number of man hours required in inspecting products. This project is a detailed analysis of several methods of computer vision and machine learning to identify stones in fabric images.**

The analysis starts with traditional feature-based methods that include Histogram of Oriented Gradients (HOG) with Support Vector Machines (SVM) and LinearSVC which proved inadequate in representing intricate texture variations. Additional testing with Convolutional Neural Networks (CNNs) indicated issues related to overfitting. To overcome these limitations, a composite framework of combining YOLO-based object detection and classifiers based on machine learning was constructed.

Different combinations were tested such as YOLO + PCA + SVM and YOLO + Multi-Layer Perceptron (MLP). The last YOLO + MLP model was more effective as it combines spatial detection with non-linear classification abilities. The report gives a stepwise description of each of the experimental steps and failed attempts, design choices, and final results.

***Index Terms*—Fabric defect detection, Object detection, YOLO, Multi-Layer Perceptron (MLP), Focal loss, Hybrid models, Computer vision, Machine learning, Deep learning, Class imbalance**

I. Introduction

The process of stone detection is one of the most important in the textile industry, as any miss count, even the smallest ones, may considerably affect the quality of the products and their economic cost. Conventional inspection processes depend on a lot of manual observation which is time consuming and can also be subject to human error.

Automated systems have become more feasible with the development of computer vision. Nevertheless, it is difficult to identify the flaws in stone detection because the texture pattern is complex and irregular.

This project will attempt to identify stones in fabric images through a systematic method and an iterative process. There were a variety of techniques tried and tested as opposed to using just one. The evolution of classical feature-based models to deep learning and mixed methods is a representation of a realistic engineering workflow, in which each one of the methods was examined, enhanced, or abandoned depending on its performance. The ultimate model was chosen even after a lot of trial and error.

II. Methodology

A. Data Preprocessing and Augmentation

To improve model performance, several preprocessing techniques were applied. Contrast enhancement was performed using Contrast Limited Adaptive Histogram Equalization (CLAHE), which significantly improved the visibility of subtle stones in fabric textures. This step helped highlight local variations that are otherwise difficult to detect.

In addition to preprocessing, data augmentation techniques were employed to increase dataset diversity and reduce overfitting. Images were rotated at multiple angles ranging from 0° to 315°, and horizontal flipping was applied. These transformations ensured that the model could generalize better to different orientations and variations of defects.

B. Classical Machine Learning Approaches

The first stage of the project was devoted to classical machine learning techniques. Image patches were used to extract Histogram of Oriented Gradients (HOG) features and mode a Support Vector Machine (SVM) classifier. Although HOG is a good edge/gradient capture procedure, it was inadequate to represent detailed fabric textures. The model had a high false positive and could not generalize well to unseen data.

LinearSVC was also tested with the aim of enhancing the efficiency of computations. It led to a decrease in training time but with no significant increases in classification performance. Linear decision boundary could not embrace the non-linear trends exhibited by the data. Consequently, the two HOG-based methods were finally abandoned.

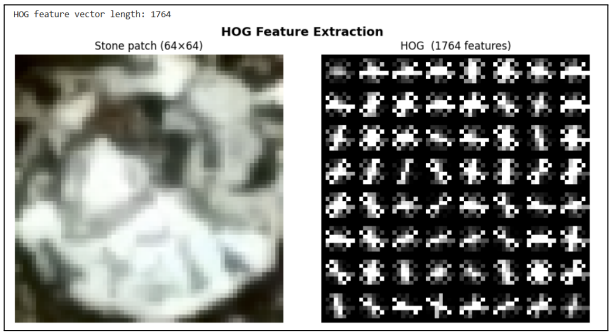

Fig. 1. Output of the HOG + SVM model illustrating high false positives and poor generalization on fabric defect detection.

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Fabric 0: P=0.722 R=0.611 F1=0.662 TP=91 FP=35 FN=58

==================================================
  OVERALL  P=0.722  R=0.611  F1=0.662
  TP=91  FP=35  FN=58
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Fig. 2. Final detection output of the proposed YOLO + MLP model showing detected stone defects with bounding boxes and confidence scores.

C. Deep Learning Approach

In the style of classical techniques, a Convolutional Neural Network (CNN) was built to automatically learn hierarchical features of images. CNNs are characterized by their capability to extract complex patterns and are hence appropriate in image tasks.

The CNN model was however very problematic because the model overfitted the training data very fast, thereby poor performance on validation samples. Stable performance was not possible even though various architectures and hyper-parameters were tested. This highlighted the importance of diversity in deep learning applications. Therefore, CNN-based approach was not chosen as the ultimate solution.

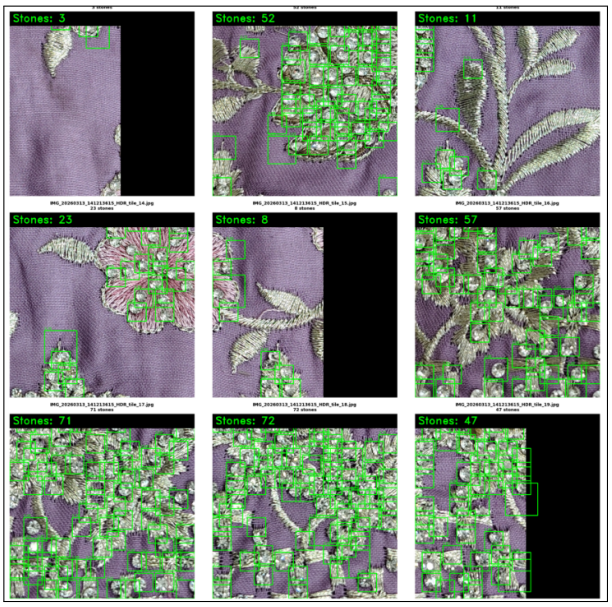

Fig. 3. Detection results using the CNN + SVM model for identifying decorative stone defects in fabric images.

D. Transition to Object Detection

To overcome the shortcomings of patch-based classification, the method changed to object detection with YOLO (You Only Look Once). YOLO is also very effective in the detection of objects in images as opposed to the past because unlike the earlier approaches, it does not require any additional steps since it does localization and classification in a single step.

The launching of YOLO was a milestone enhancement to the pipeline because it allowed the defect regions to be directly detected with spatial awareness. This did away with tiresome sliding window methods and enhanced computation speed.

E. Hybrid Model Development

Continuing the idea of YOLO, the hybrid models were created through the integration of the deep learning-based detection and the classical machine learning classifiers. The first hybrid method applied YOLO to identify regions of interest and then extracted the features and reduced the dimensions based on Principal Component Analysis (PCA). An SVM was then used to classify these reduced features.

Though this method portrayed moderate improvement, PCA led to loss of significant information, which influenced the accuracy of classification. In order to eliminate this constraint, a Multi-Layer Perceptron (MLP) was used instead of the SVM classifier. The MLP, which models non-linear relationships, was more adaptable to complex texture patterns.

The ultimate model pipeline involved YOLO to detect the regions, followed by the extraction of features and the classification with the help of MLP. This fused solution was able to effectively integrate space detection and powerful classification, leading to enhanced functionality.

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Fabric 0: P=0.568 R=0.557 F1=0.563 TP=83 FP=63 FN=66
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Fig. 4. Detection results using the YOLO + PCA + RBF-SVM model.

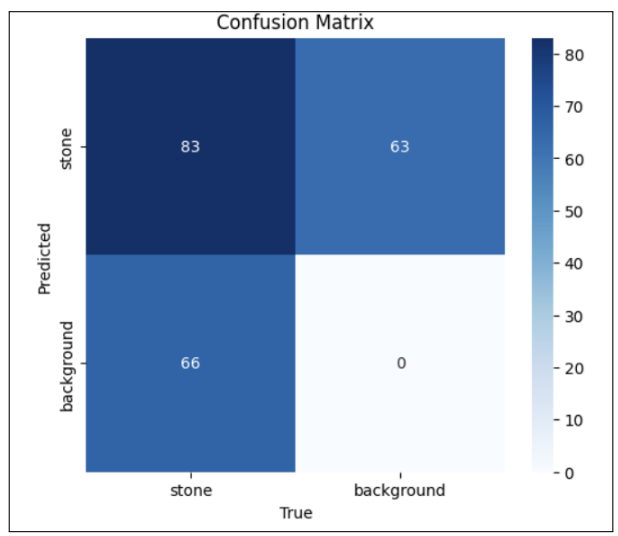

Fig. 5. Detection results using the YOLO + PCA + RBF-SVM model.

III. Results

Several experiments were performed to evaluate the effectiveness of the proposed hybrid detection framework combining YOLO with an MLP classifier trained using Focal Loss.

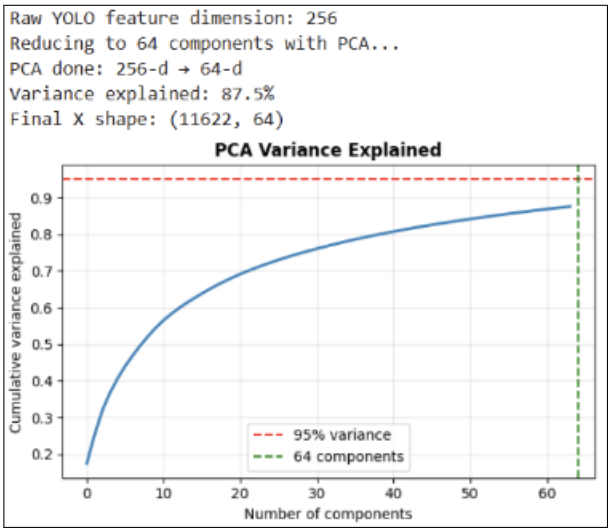

Fig. 6. Cumulative variance explained by Principal Component Analysis (PCA) applied to YOLO-extracted features, illustrating dimensionality reduction from 256 to 64 components while preserving 87.5% of the original feature information for RBF-SVM classification.

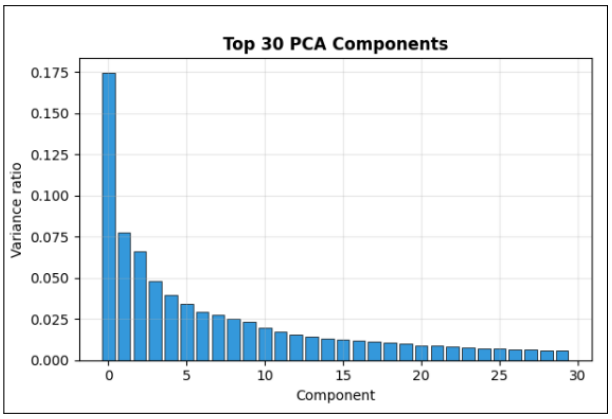

Fig. 7. Explained variance ratio of the top PCA components derived from YOLO features.

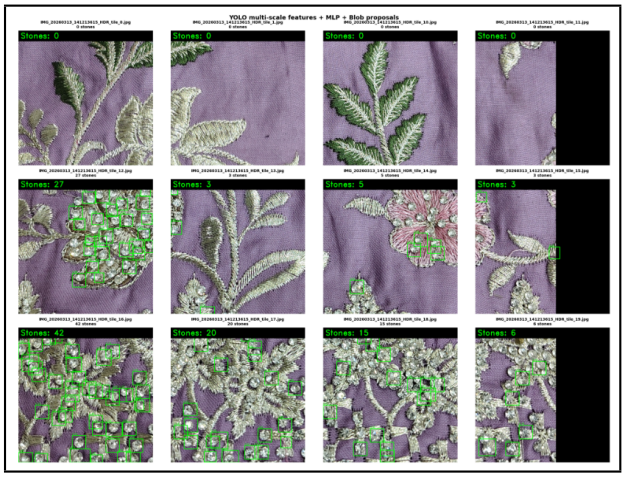

Fig. 8. Final detection output of the proposed YOLO + MLP model showing detected stone defects with bounding boxes and confidence scores.

TABLE I
FINAL MODEL PERFORMANCE (YOLO + MLP + FOCAL LOSS)

Metric	Value
Precision	0.722
Recall	0.611
F1 Score	0.662
True Positives (TP)	91
False Positives (FP)	35
False Negatives (FN)	58

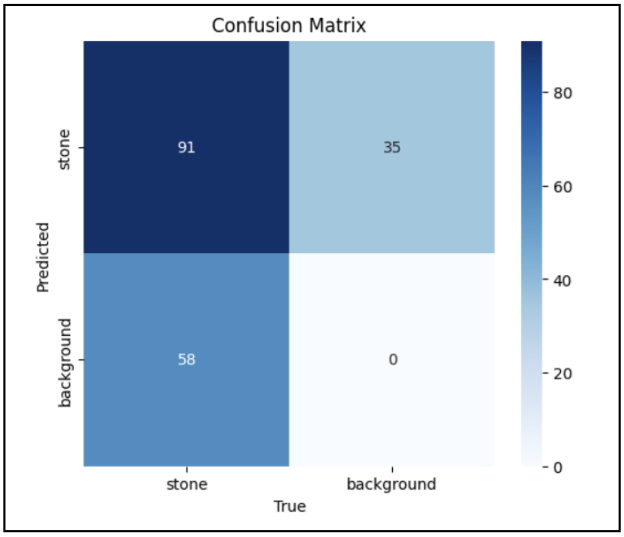

Fig. 9. Final detection output of the proposed YOLO + MLP model showing detected stone defects with bounding boxes and confidence scores.

The evaluation of the proposed hybrid system consisting of multi-scale YOLO for feature extraction combined with the MLP-based classifier trained via Focal Loss was performed on labeled fabric data. Specifically, a precision rate of 0.722, recall rate of 0.611, and F1-score of 0.662 were obtained, with 91 true positives, 35 false positives, and 58 false negatives identified. This shows that the system successfully detects most of the stone defects despite a slight imbalance between precision and recall rates due to complex fabric textures.

The performance improvement can be explained through a combination of the usage of multi-scale feature representations and the MLP classifier, which learns non-linear decision boundaries. Furthermore, Focal Loss addresses class imbalance issues by focusing on difficult samples, while the hybrid proposal generation method improves efficiency by reducing unnecessary candidate regions. Even though some defects remain undetected due to clustering and occlusions, the overall approach demonstrates effective performance in the detection task.

$$FL(p) = -\alpha(1-p)^{\gamma}\log(p) \tag{1}$$

```
Fabric 0: P=0.722 R=0.611 F1=0.662 TP=91 FP=35 FN=58

==================================================
  OVERALL  P=0.722  R=0.611  F1=0.662
  TP=91  FP=35  FN=58
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Fig. 10. Final detection output of the proposed YOLO + MLP model showing detected stone defects with bounding boxes and confidence scores.

The proposed system was able to detect defect regions with improved robustness compared to earlier approaches. While the recall indicates that some defects are still missed, the overall performance validates the effectiveness of combining deep learning-based feature extraction with a non-linear classifier and optimized loss function.

IV. Discussion

The experimental results highlight the progressive improvement achieved by transitioning from traditional feature-based methods to hybrid deep learning approaches. The initial HOG + SVM model struggled to capture complex fabric textures, resulting in high false positives and poor generalization. Similarly, CNN-based feature extraction combined with SVM improved representation capability but was still limited by the linear decision boundaries of the classifier. The introduction of YOLO significantly enhanced performance by enabling effective region proposal, allowing the model to focus on relevant areas rather than processing the entire image exhaustively.

Further improvements were observed with hybrid models, particularly the integration of multi-scale YOLO features with an MLP classifier trained using Focal Loss. This combination enabled the system to learn non-linear feature representations while addressing class imbalance, leading to better precision and overall robustness. Although the YOLO + PCA + RBF-SVM approach improved efficiency, it suffered from information loss due to dimensionality reduction. In contrast, the final YOLO + MLP model demonstrated a more balanced performance, though some limitations remain in detecting densely clustered or occluded stones. Overall, the results confirm that combining deep feature extraction with non-linear classification provides a more effective solution for fabric stone detection.

V. Conclusion

This work explored multiple approaches for fabric stone detection, starting from classical methods such as HOG + SVM and CNN + SVM, which showed limited performance due to their inability to effectively capture complex textures. The introduction of YOLO improved region-based detection, while hybrid models combining YOLO with PCA and RBF-SVM provided moderate gains but suffered from information loss due to dimensionality reduction.

The final proposed approach, integrating multi-scale YOLO features with an MLP classifier trained using Focal Loss, achieved the best performance by learning non-linear feature representations and handling class imbalance effectively. The model demonstrated a balanced detection capability with an F1-score of 0.662. Although some limitations remain in detecting densely clustered stones, the overall results validate the effectiveness of the proposed hybrid framework and suggest potential for further improvement.

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